

Cairo University

Faculty of Computers and Artificial Intelligence



CS251

Intro. to Software Engineering

Software Requirements Specifications

Version 1.0

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Project: Masroofy

Software Requirements Specifications

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Document Purpose and Audience

Purpose: This document specifies the software requirements for our project to ensure all stakeholders and developers understand the system's functionality and constraints.

Audience: The primary audience includes the Course Instructor (Dr. M. El-Ramly), the TA, and the development team.

Introduction

Software Purpose

The purpose of this software is to provide an easy-use, offline, manual micro-budgeting mobile application designed specifically for Egyptian students and young adults. It helps replace traditional tracking methods like physical notebooks or basic note app.

Software Scope

The scope of this project is an offline Android-based mobile application.

features included:

- **Cycle Management Module:** A setup system where users input their cash allowance and define the timeframe.
- **Expense Tracking System:** manual entry interface allowing users to rapidly log daily transactions and assign them to specific lifestyle categories.



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- **Financial Dashboard:** A visual interface providing real-time summaries of remaining funds, the current daily limit, and visual charts (e.g., pie charts) breaking down spending by category.
- **Alerting System:** A local push notification system that warns the user when 80% of their total cycle allowance has been consumed.

Definitions, acronyms, and abbreviations

Term / Acronym	Definition
Micro-Budgeting	The practice of managing small, frequent cash flows over a short, predefined time cycle (e.g., a week or a month), tailored to daily living expenses.
Allowance	The total starting amount of money available for the user to spend during the current active cycle.
Safe Daily Limit	A dynamically recalculated value indicating exactly how much cash the user can spend <i>today</i> without running out of money before the cycle ends.
EGP	Egyptian Pound; the standard local currency used for all calculations within the application.
UI/UX	User Interface / User Experience; refers to the visual design, layout, and overall ease-of-use of the application.



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API	Application Programming Interface; a mechanism that allows two software components to communicate. (Note: This software intentionally avoids banking APIs to maintain an offline state).
SQLite	A lightweight, local database engine used to securely store the user's financial transactions directly on their mobile device without needing internet access.
MVC	Model-View-Controller; the underlying architectural software pattern used to structure the application's code.



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Requirements

Functional Requirements

ID	Requirement Name	Requirement Description
FR-1	Allowance Initialization	The system shall allow the user to initialize an allowance cycle by inputting a total cash amount and selecting a specific start and end date.
FR-2	Limit Calculation	The system shall automatically calculate the initial "Safe Daily Limit" by dividing the total allowance by the total number of days in the defined cycle.
FR-3	Expense Logging	The system shall provide a logging interface where users can record an expense by entering an amount and selecting a category (e.g., Food, Transport).
FR-4	Dynamic Rollover	The system shall dynamically recalculate the daily limit for remaining days whenever an expense is logged or when unspent funds from the previous day are rolled over.



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FR-5	Spending Dashboard	The system shall generate a visual dashboard featuring a pie chart that displays the percentage of total spending allocated to each category.
FR-6	Usage Alerts	The system shall monitor the total budget and trigger a local push notification alert once the user has consumed 80% of their total allowance.
FR-7	Local Persistence	The system shall ensure all transaction data is immediately persisted to a local SQLite database after every user entry.



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Non Functional Requirements

Category	Details
Performance	• The expense logging process must be completed in 3 taps or fewer. • Dashboard updates and calculations must occur in less than 1 second.
Availability	• The application must be 100% functional offline, requiring no internet connection for core features or data entry.
Security/Privacy	• All financial data must be stored exclusively on the user's local device; the system shall not transmit data to any external server or API.
Reliability	• The system shall maintain data integrity in the SQLite database during unexpected application terminations or device restarts.



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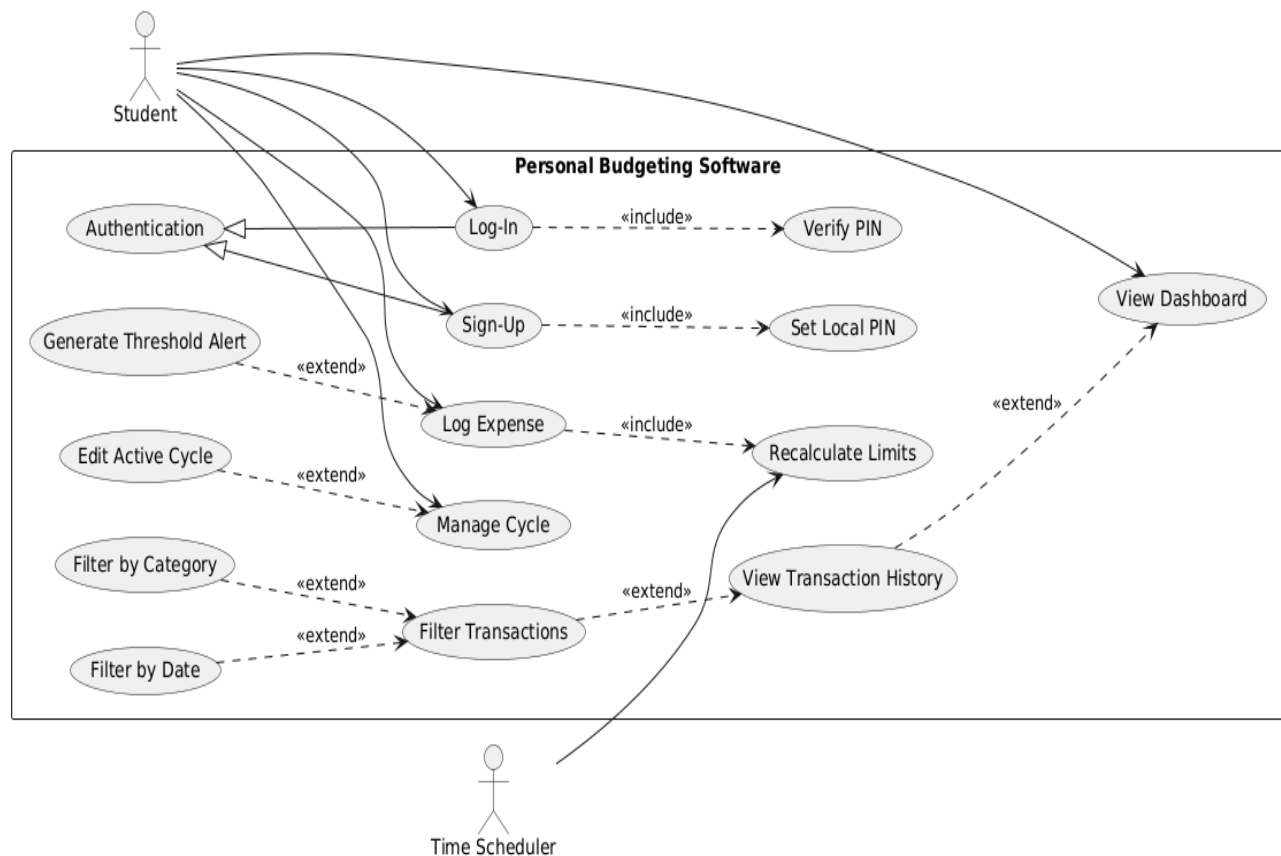
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Usability	• The interface shall be designed for one-handed use, specifically targeting rapid entry for students on the move.
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System Models

Use Case Model





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Enriched User Stories

- User Story #1

User Story ID	US #1
User Story Name	Set Initial Budget Cycle
Actors	College Student
Description	As a college student, I like to be able to input my total cash allowance and the duration of my spending cycle So the system can calculate my starting daily limit and track my progress.
Pre condition	The application is installed, and the user has no active budget cycle running.
Post condition	A new budget cycle is created in the SQLite database, and the "Safe Daily Limit" is calculated and displayed.
Acceptance Criteria	Given I am on the setup screen for the first time When I enter a valid numeric amount (ex: 3000 EGP), select a start date (ex: today), and select an end date (ex: 30 days from now) Then the system saves this configuration and navigates me to the dashboard

- Scenarios

Normal Scenario

Actor	
Action	System Response



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User opens the app for the first time.	System detects no active cycle and displays "Initial Setup" screen.
User enters total Budget amount (ex: 1000).	
User selects start and end dates for the cycle.	System validates date range.
Click Start Cycle .	System calculates daily limit and saves data to SQLite.
	System displays dashboard.

Exceptional Scenario

Actor	Action	System Response
	User enters negative number or zero in allowance field.	
	Click Start Cycle .	Input is invalid.
		System rejects entry and displays error message: "Allowance must be a positive number."

- **Screen Design**



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3-Intialization

- **Data Dictionary:**

Element Label	Type/Length	Data Validation / Business Rule
Total Allowance	Decimal / Float	Must be > 0; cannot be empty.
Start Date	Date Picker	Defaults to current device date.
End Date	Date Picker	Must be strictly greater than Start Date.

- **User Story #2**

User Story ID	US #2
User Story Name	Rapid Expense Logging
Actors	College Student
Description	As a college student I like to be able to record an expense in under three taps



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	So I can track my spending instantly without manual friction.
Pre condition	An active budget cycle exists in the system.
Post condition	The transaction is saved to the SQLite database; the "Safe Daily Limit" and dashboard charts are updated.
Acceptance Criteria	Given I am on the Quick-Entry screen When I fill in the "Amount" field and tap a "Category" icon (ex: Food) Then the system saves the record and redirects me to the dashboard.

- Scenarios**

Normal Scenario

Actor		System Response
Action		
	User enters the expense amount (ex: 50).	
	User taps the “ Transport ” category icon.	
	Click Save .	System validates numeric data.
		System writes the transaction to local SQLite database.
		System recalculates daily limit and updates charts.
		System returns to Dashboard with “ Saved ” confirmation.

Exceptional Scenario

Actor		System Response
Action		
	User enters non-numeric text in amount field.	
	Click Save .	Data type is invalid.



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System rejects entry and displays error message: **“Please enter a valid number.”**

- **Screen Design**

5-New Expense

✕

Amount

✓

30 EGP

Select Category

Food

Transport

Entertainment

Other

Add Category

- **Data Dictionary:**

Element Label	Type/Length	Data Validation / Business Rule
Expense Amount	Decimal / Float	Must be numeric; maximum 10 digits.
Category Icon	Selection (ID)	User must select exactly one from the grid.
Timestamp	DateTime	Auto-generated by system at moment of saving.

- **User Story #3**



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User Story ID	US #3
User Story Name	Dynamic Daily Limit View
Actors	College Student
Description	As a college student, I like to be able to see my updated "Safe Daily Limit" on the home screen, So I can immediately know how much I am allowed to spend today without running out of money before the cycle ends.
Pre condition	An active budget cycle has been initialized.
Post condition	The current "Safe Daily Limit" is displayed prominently on the dashboard.
Acceptance Criteria	Given I have logged expenses or passed a day without spending, When I open the app's dashboard, Then the system displays a recalculated daily limit based on the remaining balance and remaining days.

- Scenarios**

Normal Scenario

Actor	Action	System Response
	1- User opens the application.	2- System retrieves the remaining balance and remaining days from SQLite.
		3- System calculates the new limit: Remaining Balance / Remaining Days.
		4- System displays the " Safe Daily Limit " in a large, central font on the dashboard.



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Software Requirements Specifications

Exceptional Scenario

Actor	Action	System Response
	1- User opens the app on the final day of the cycle.	2- System calculates the limit for the last 24 hours.
		3- System displays a " Final Day " warning badge alongside the limit.

● Screen Design



● Data Dictionary:

Element Label	Type/Length	Data Validation / Business Rule
Safe Daily Limit	Decimal / Float	Calculated: Remaining Budget ÷ Remaining Days.
Category Totals	Decimal / Float	Aggregated sum of transactions per category.



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Software Requirements Specifications

Element Label	Type/Length	Data Validation / Business Rule
Pie Chart Slice	Graphic Percentage	Represents $(\text{Category Total} \div \text{Total Spent}) \times 100$.

● User Story #4

User Story ID	US #4
User Story Name	Visual Spending Insights
Actors	College Student
Description	As a college student, I like to be able to view a visual breakdown of my expenses by category So I can identify where I am spending the most money and adjust my habits accordingly.
Pre condition	The user has logged at least one expense in the current budget cycle.
Post condition	A pie chart is rendered on the dashboard reflecting current spending percentages.
Acceptance Criteria	Given I am on the Dashboard screen, When the system calculates transaction totals for each category, Then a pie chart is displayed with segments representing each category's share of total spending.

● Scenarios

Normal Scenario



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Software Requirements Specifications

Actor Action	System Response
1- User navigates to the Dashboard.	2- System queries SQLite for all transactions in the current cycle.
	3- System aggregates totals per category (e.g., Food, Transport).
	4- System generates and displays a pie chart where each slice represents a category's percentage of total expenditure.

Exceptional Scenario

Actor Action	System Response
1- User navigates to the Dashboard with zero expenses logged.	2- System identifies that the transaction table is empty for the current cycle.
	3- System displays a placeholder message: "No data available. Log an expense to see your insights."

- Screen Design



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Software Requirements Specifications



• Data Dictionary:

Element Label	Type/Length	Data Validation / Business Rule
Safe Daily Limit	Decimal / Float	Calculated: Remaining Budget ÷ Remaining Days.
Category Totals	Decimal / Float	Aggregated sum of transactions per category.
Pie Chart Slice	Graphic Percentage	Represents (Category Total ÷ Total Spent) × 100.

• User Story #5

User Story ID	US #5
User Story Name	Daily Rollover Management
Actors	College Student



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Description	As a college student, I like to be able to have my unspent daily limit automatically added to the remainder of my cycle, So that my daily limit for the following days increases as a reward for saving.
Pre condition	A new day has started, and the user spent less than their allocated "Safe Daily Limit" on the previous day.
Post condition	The "Safe Daily Limit" for all remaining days in the cycle is recalculated and increased in the SQLite database.
Acceptance Criteria	Given my daily limit was 50 EGP and I spent only 30 EGP, When the system clock reaches 00:00 (midnight), Then the remaining 20 EGP is redistributed across the remaining days of the cycle, increasing the displayed daily limit.

- **Scenarios**

Normal Scenario

Actor	Action	System Response
	1- User opens the app on a new day.	2- System checks the timestamp of the last login and the current balance.
		3- System identifies unspent funds from the previous day.
		4- System divides the Total Remaining Balance by the Number of Remaining Days (including today) .
		5- System updates the dashboard with the newly increased " Safe Daily Limit. "



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Software Requirements Specifications

Exceptional Scenario

Action	Actor	System Response
1- User opens the app after overspending the previous day's limit.		2- System identifies a " negative rollover " (deficit).
		3- System recalculates the limit, which results in a lower daily amount for the remaining days.
		4- System displays the updated limit with a subtle color change (e.g., orange) to indicate a tighter budget.

- Screen Design



Redistribution and the new safe limit is displayed here (Dashboard)

- Data Dictionary:



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Software Requirements Specifications

Element Label	Type/Length	Data Validation / Business Rule
Safe Daily Limit	Decimal / Float	Calculated: Remaining Budget ÷ Remaining Days.
Category Totals	Decimal / Float	Aggregated sum of transactions per category.
Pie Chart Slice	Graphic Percentage	Represents (Category Total ÷ Total Spent) × 100.

• User Story #6

User Story ID	US #6
User Story Name	Budget Threshold Notification
Actors	College Student
Description	As a college student, I like to be able to receive a local push notification when I have spent 80% of my total budget, So I can make a plan to prevent running out of money before the end of the week or month.
Pre condition	The user has an active allowance cycle and has logged multiple expenses.
Post condition	A system notification is triggered and displayed on the user's device.
Acceptance Criteria	Given my initial allowance was 1000 EGP, When I log a transaction that brings my total spending to 800 EGP (80%) or more, Then the system triggers a local notification saying: "Warning: You have used 80% of your allowance."



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Software Requirements Specifications

- Scenarios

Normal Scenario

Actor Action	System Response
1- User enters an expense amount that crosses the 80% threshold of the total cycle budget.	2- System saves the transaction to the SQLite database.
	3- System calculates the new total expenditure percentage.
	4- System identifies that the total is $\geq 80\%$ of the initial budget.
	5- System triggers a local push notification as an early warning.

Exceptional Scenario

Actor Action	System Response
1- User logs an expense that immediately jumps from 70% to 100% of the budget.	2- System detects that both the 80% and 100% thresholds were crossed.
	3- System prioritizes the most critical alert .
	4- System sends a " Budget Exhausted " notification instead of the 80% warning .

- Screen Design



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Software Requirements Specifications



Alerts are displayed here (Dashboard)

● Data Dictionary:

Element Label	Type/Length	Data Validation / Business Rule
Filter Selection	String / Dropdown	Matches existing categories in SQLite database.
PIN Field	Integer / 4 digits	Must match the stored hashed PIN in local storage.
Reset Confirmation	Boolean / Binary	Requires explicit user "Confirm" before deletion.

● User Story #7

User Story ID	US #7
User Story Name	Transaction History Review
Actors	College Student



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Software Requirements Specifications

Description	As a college student, I like to be able to view a chronological list of all my recorded expenses, So I can track exactly when and where my money was spent during the current cycle.
Pre condition	The user has an active budget cycle and has logged at least one transaction.
Post condition	A list of transactions is displayed on the screen, sorted by date and time.
Acceptance Criteria	Given I have logged several expenses (e.g., Food and Transport), When I navigate to the "History" tab or screen, Then the system displays a list showing each expense amount, category, and timestamp.

- **Scenarios**

Normal Scenario

Actor	Action	System Response
	1- User taps the "History" icon on the bottom navigation bar.	2- System queries the local SQLite database for all transactions linked to the current cycle ID.
		3- System sorts the transactions in descending order (newest first) .
		4- System renders a list where each row shows the category icon, date, and amount spent .

Exceptional Scenario



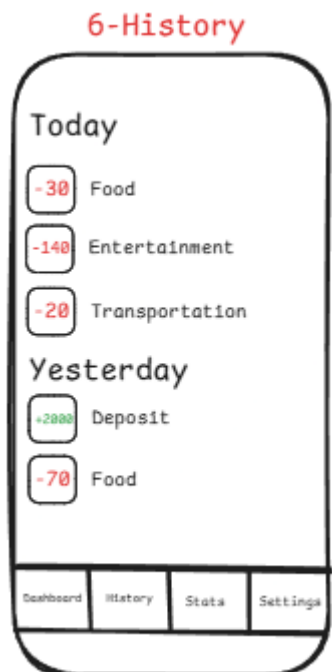
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Software Requirements Specifications

Actor Action	System Response
1- User navigates to the History screen with no transactions logged yet.	2- System identifies that the SQLite transaction table is empty for the current cycle .
	3- System displays a " No Transactions Found " placeholder message.
	4- System provides a shortcut button to " Log your first expense. "

- Screen Design



- Data Dictionary:

Element Label	Type/Length	Data Validation / Business Rule
Filter Selection	String / Dropdown	Matches existing categories in SQLite database.



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Software Requirements Specifications

Element Label	Type/Length	Data Validation / Business Rule
PIN Field	Integer / 4 digits	Must match the stored hashed PIN in local storage.
Reset Confirmation	Boolean / Binary	Requires explicit user "Confirm" before deletion.

• User Story #8

User Story ID	US #8
User Story Name	Edit or Delete Transaction
Actors	College Student
Description	As a college student, I like to be able to modify or remove a previously recorded expense, So I can correct input mistakes and keep my daily limit calculations accurate.
Pre condition	The user has at least one recorded expense in the current cycle.
Post condition	The transaction is updated or removed from the SQLite database, and the "Safe Daily Limit" is automatically recalculated.
Acceptance Criteria	Given I am on the Transaction History screen, When I select an expense to "Edit" (change amount/category) or "Delete", Then the system updates the database and immediately reflects the new limit on the dashboard.

• Scenarios



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Software Requirements Specifications

Normal Scenario

Action	Actor	System Response
1- User opens the History screen and taps the "Edit" icon next to a 50 EGP entry.		2- System displays the entry details in an editable form .
3- User changes the amount to 30 EGP and clicks Save .		4- System validates the new numeric data .
		5- System updates the record in the local SQLite database .
		6- System triggers the calculation engine to update the "Safe Daily Limit" .
		7- System returns to History with a "Transaction Updated" confirmation.

Exceptional Scenario

Action	Actor	System Response
1- User selects "Delete" on a transaction.		2- System prompts: "Are you sure you want to delete this? This will update your daily limit."
3- User confirms deletion.		4- System removes the record from SQLite .
		5- System recalculates the increased daily limit for the remaining days.

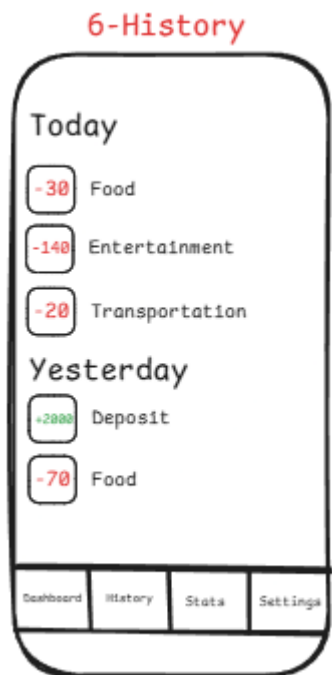
- **Screen Design**



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Software Requirements Specifications



- **Data Dictionary:**

Element Label	Type/Length	Data Validation / Business Rule
Filter Selection	String / Dropdown	Matches existing categories in SQLite database.
PIN Field	Integer / 4 digits	Must match the stored hashed PIN in local storage.
Reset Confirmation	Boolean / Binary	Requires explicit user "Confirm" before deletion.

- **User Story #9**

User Story ID	US #9
User Story Name	Filter Transaction History
Actors	College Student



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Software Requirements Specifications

Description	As a college student, I like to be able to filter my transactions by specific categories or date ranges, So I can analyze my spending for specific items like "Food" or review what I spent during a particular weekend.
Pre condition	The user is on the Transaction History screen with multiple logged expenses.
Post condition	The displayed list is filtered to show only transactions matching the selected criteria.
Acceptance Criteria	Given I am on the History screen, When I select "Transport" from the category filter, Then the system hides all other expenses and shows only Transport-related logs.

- Scenarios**

Normal Scenario

Action	Actor	System Response
1- User taps the "Filter" icon on the History screen.		2- System displays a dropdown with available categories and a date-range picker .
3- User selects "Food" and sets the date to "Yesterday."		4- System queries the SQLite database using WHERE clauses for category and date.
		5- System updates the UI to display only the matching transactions.

Exceptional Scenario



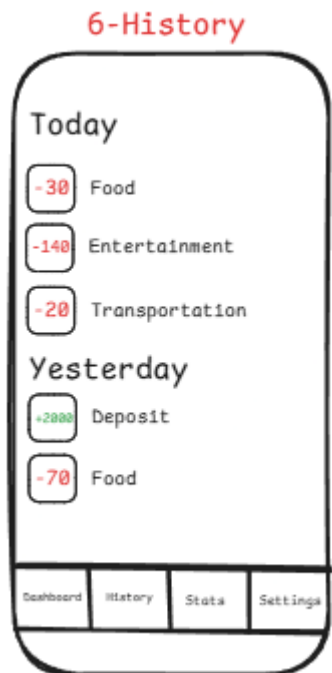
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Software Requirements Specifications

Actor	Action	System Response
	1- User applies a filter for a category with no logged transactions.	2- System performs the search and finds zero results.
		3- System displays a message: " No transactions found for the selected filter. "

- Screen Design



- Data Dictionary:

Element Label	Type/Length	Data Validation / Business Rule
Filter Selection	String / Dropdown	Matches existing categories in SQLite database.
PIN Field	Integer / 4 digits	Must match the stored hashed PIN in local storage.



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Software Requirements Specifications

Element Label	Type/Length	Data Validation / Business Rule
Reset Confirmation	Boolean / Binary	Requires explicit user "Confirm" before deletion.

● User Story #10

User Story ID	US #10
User Story Name	Offline Local Data Persistence
Actors	College Student
Description	As a college student, I like to be able to use all application features while my device is offline, So I can manage my cash allowance anywhere regardless of internet connectivity or cellular signal.
Pre condition	The device has no active internet or network connection.
Post condition	All inputs are saved to the local SQLite database and are available upon the next application launch.
Acceptance Criteria	Given my phone is in airplane mode, When I log an expense or view my dashboard, Then the system performs all calculations and saves the data locally without any "no connection" errors.

● Scenarios

Normal Scenario

Actor	Action	System Response
	1- User opens the app while in an area with zero internet coverage.	2- System launches immediately without attempting an external login or API sync.



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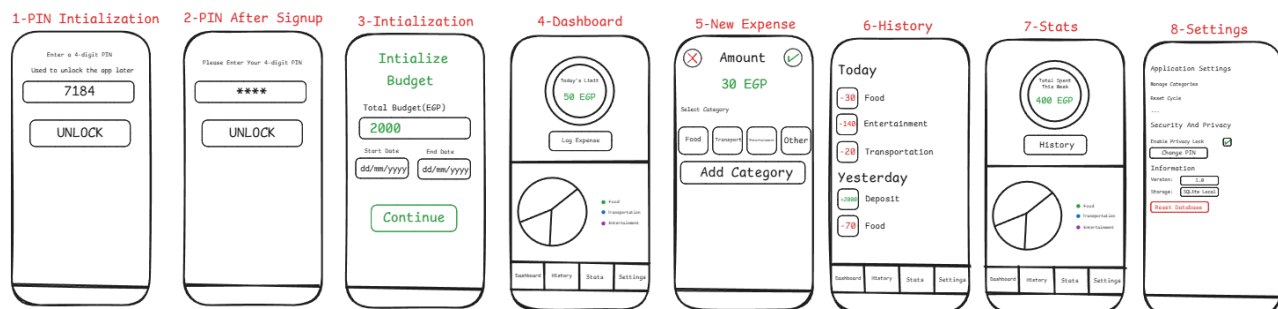
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3- User logs an expense of 40 EGP.	4- System writes the entry directly to the local SQLite file.
	5- System updates the "Safe Daily Limit" using local logic.
	6- System displays a "Saved Locally" confirmation

Exceptional Scenario

Actor	Action	System Response
	1- User restarts the device while the app is running in an offline state.	2- System triggers the SQLite journaling mechanism to prevent data corruption.
	3- User reopens the app after the restart.	4- System restores the state from the last committed transaction in SQLite.
		5- System confirms that no data was lost during the power cycle.

• Screen Design



• User Story #11

User Story ID	US #11
User Story Name	Cycle Reset and Data Clearance
Actors	College Student
Description	As a college student,



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	I like to be able to end my current budget cycle early and clear all associated transaction data, So I can start a fresh cycle immediately if my financial circumstances change or a new allowance arrives early.
Pre condition	The user has an active budget cycle and is in the "Settings" or "History" menu.
Post condition	The active cycle and all related transactions are deleted from the SQLite database, and the user is returned to the "Initial Setup" screen.
Acceptance Criteria	Given I am on the Settings screen, When I select "Reset Current Cycle" and confirm the deletion, Then the system wipes the local budget data and prompts me to enter a new allowance.

- Scenarios**

Normal Scenario

Actor	Action	System Response
	1- User navigates to Settings and taps "Reset Current Cycle."	2- System displays a confirmation dialog: "This will permanently delete all logs for this cycle. Continue?"
	3- User taps "Yes, Reset."	4- System executes a DELETE command on the cycle and transaction tables in SQLite .
		5- System clears the "Safe Daily Limit" from the local cache.
		6- System redirects the user to the "Allowance Initialization" screen.

Exceptional Scenario



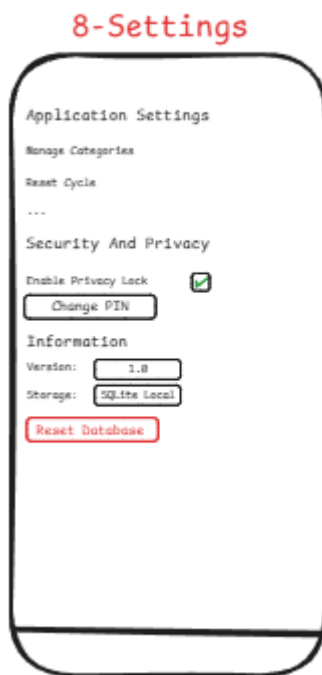
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Software Requirements Specifications

Actor	Action	System Response
	1- User taps "Reset" by accident.	2- System displays the confirmation dialog .
	3- User taps "Cancel" or anywhere outside the dialog.	4- System closes the dialog and makes no changes to the database.
		5- System returns the user to the previous screen with data intact .

- Screen Design



- Data Dictionary:

Element Label	Type/Length	Data Validation / Business Rule
Filter Selection	String / Dropdown	Matches existing categories in SQLite database.
PIN Field	Integer / 4 digits	Must match the stored hashed PIN in local storage.



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Software Requirements Specifications

Element Label	Type/Length	Data Validation / Business Rule
Reset Confirmation	Boolean / Binary	Requires explicit user "Confirm" before deletion.

● User Story #12

User Story ID	US #12
User Story Name	Local Privacy Lock
Actors	College Student
Description	As a college student, I like to be able to enable a local PIN or biometric lock, So I can ensure my personal financial records and allowance data remain private from others who might access my phone.
Pre condition	The user has initialized a budget cycle and is in the "Settings" menu.
Post condition	A local authentication flag is set in the SQLite database, requiring verification upon next app launch.
Acceptance Criteria	Given I have enabled the "Privacy Lock" in settings, When I close and then reopen the application, Then the system prompts for my device PIN or biometric (fingerprint/face) before displaying the dashboard.

● Scenarios

Normal Scenario

Actor	System Response
Action	



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Software Requirements Specifications

1- User navigates to Settings and toggles "Enable Privacy Lock."	2- System requests the user to define a 4-digit PIN or confirm biometric authentication .
3- User enters PIN and confirms.	4- System saves the encrypted hash of the PIN to the local SQLite database .
5- User closes the app and reopens it.	6- System detects the active lock and displays the authentication screen .
7- User provides valid authentication.	8- System grants access and displays the dashboard .

Exceptional Scenario

Action	Actor	System Response
1- User provides an incorrect PIN three times .		2- System identifies the failed authentication attempts .
		3- System enforces a 30-second lockout period to prevent brute-force entry.
		4- System displays a "Try again later" message.

- **Screen Design**



CS251: Phase 1 – No Fs

Project: Masroofy

Software Requirements Specifications

1-PIN Initialization

Enter a 4-digit PIN
Used to unlock the app later

7184

UNLOCK

2-PIN After Signup

Please Enter Your 4-digit PIN

UNLOCK

- Data Dictionary:**

Element Label	Type/Length	Data Validation / Business Rule
Filter Selection	String / Dropdown	Matches existing categories in SQLite database.
PIN Field	Integer / 4 digits	Must match the stored hashed PIN in local storage.
Reset Confirmation	Boolean / Binary	Requires explicit user "Confirm" before deletion.

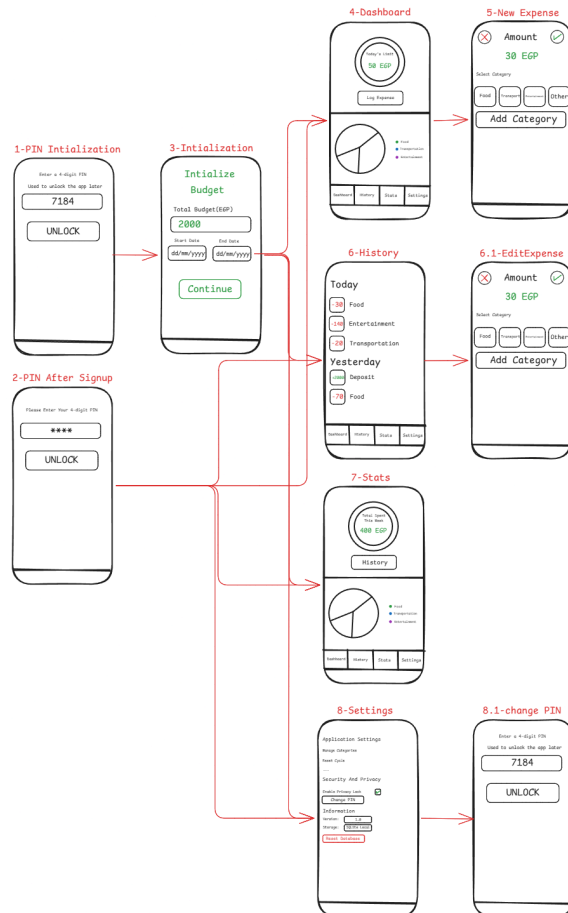
System Navigation Map



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Software Requirements Specifications



NOTE: Zoom in for details or click the link below:

<https://excalidraw.com/#json=8lUn8Czbxi-MvtnLdoPgS.R8uJC5NtdbJGTQrreAYYDw>

Tools

- PlantUML
- Excalidraw
- Figma



CS251: Phase 1 – No Fs

Project: Masroofy

Software Requirements Specifications

Ownership Report

Student	Items he created
Yamen Ali Fathi	Use case model/ mock up system navigation/design
Ziad Alaa Mokhtar	<i>introduction/ user stories</i>
Omar Mohamed Abdelkader	System Navigation Map/ user stories
Mohamed Basheir Mostafa	Requirements / <i>user stories</i>